

Simulating Safe Bite Transfer in Robot-Assisted Feeding with a Soft Head and Articulated Jaw



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1 Introduction

- Robot-assisted feeding requires close physical interaction with sensitive facial areas
- Safe bite transfer is challenging to study due to safety risks and lack of detailed physical human-robot interaction simulators [1,2]
- Need for simulation tools to evaluate safety before real-world trials

Goal: Develop a soft-body head model to enable evaluation of bite transfer safely in simulation, to personalise bite transfer parameters to the user for maximum comfort.

2 Head Modeling

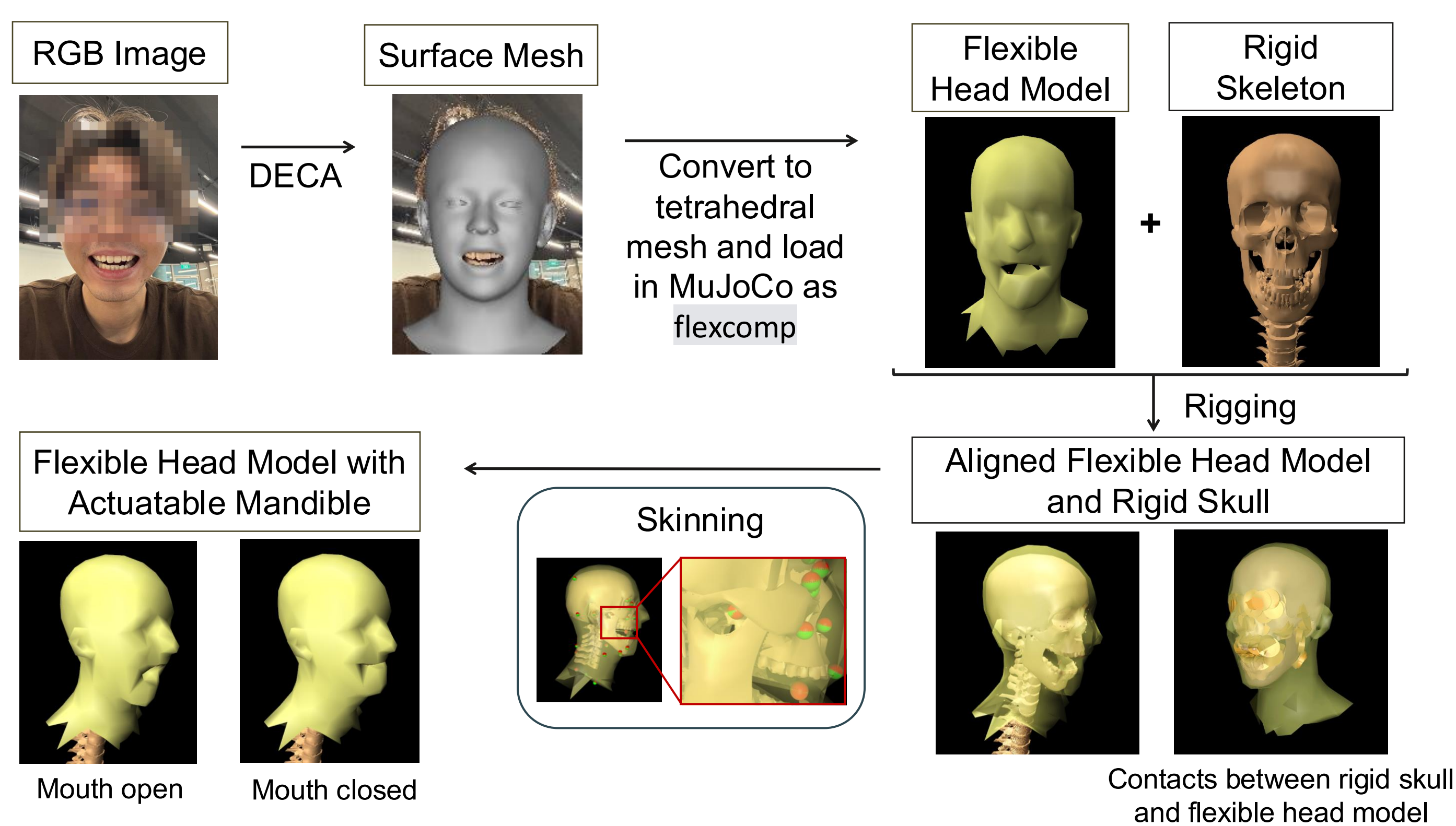


Figure 1: (Top) A flexible head model is generated from an RGB image. (Bottom) The flexible head model is combined with a rigid skeleton through rigging and skinning. Tendon elements connect the rigid and flexible models, based on sites created at contact points.

3 Bite Transfer Simulation

Phase	Parameter	Range	Increment
Entry	Entry Angle (α)	80° to 110°	10°
	Insertion Depth (d)	50 mm to 100 mm	10 mm
Exit	Exit Angle (β)	80° to 120°	10°
	Exit Depth (e)	0 mm to 40 mm	10 mm

Metrics

- Peak Contact Force:** affects instantaneous user comfort, as high forces in short duration can lead to injury
- Total Contact Force:** affects overall comfort, as prolonged low-level forces can still cause discomfort.

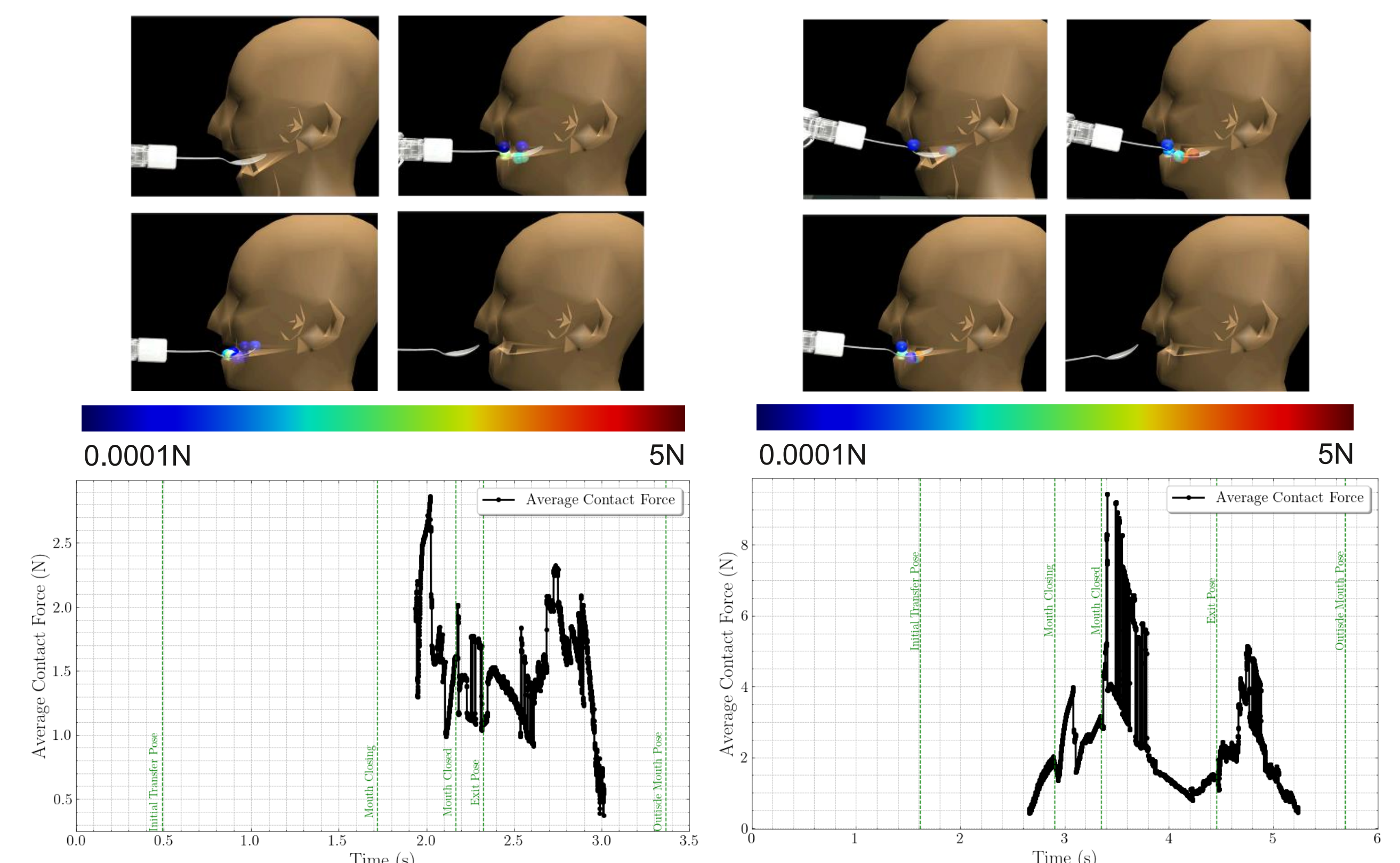


Figure 2: (Left) Optimal parameters (Right) Suboptimal parameters. (Top) Screen captures of the bite transfer process, contact forces visualized as coloured spheres. (Bottom) Contact forces over time.

4 Results

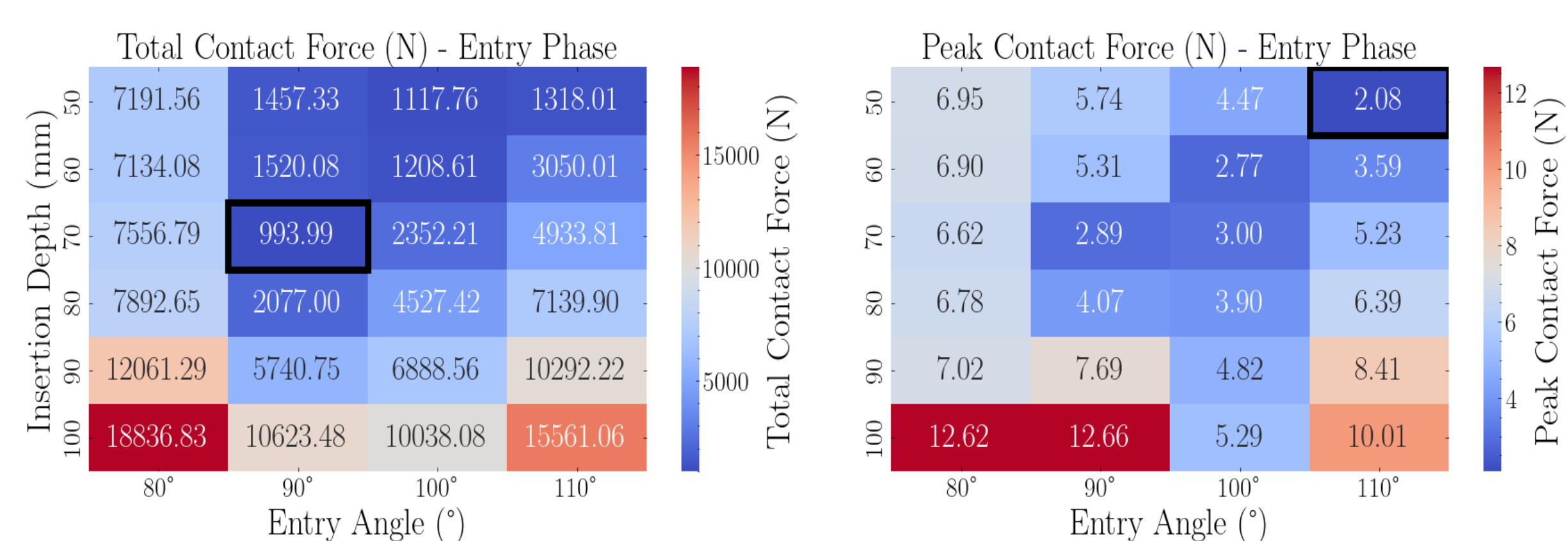


Figure 3: (Left) Total contact force and (Right) Peak contact force during entry phase

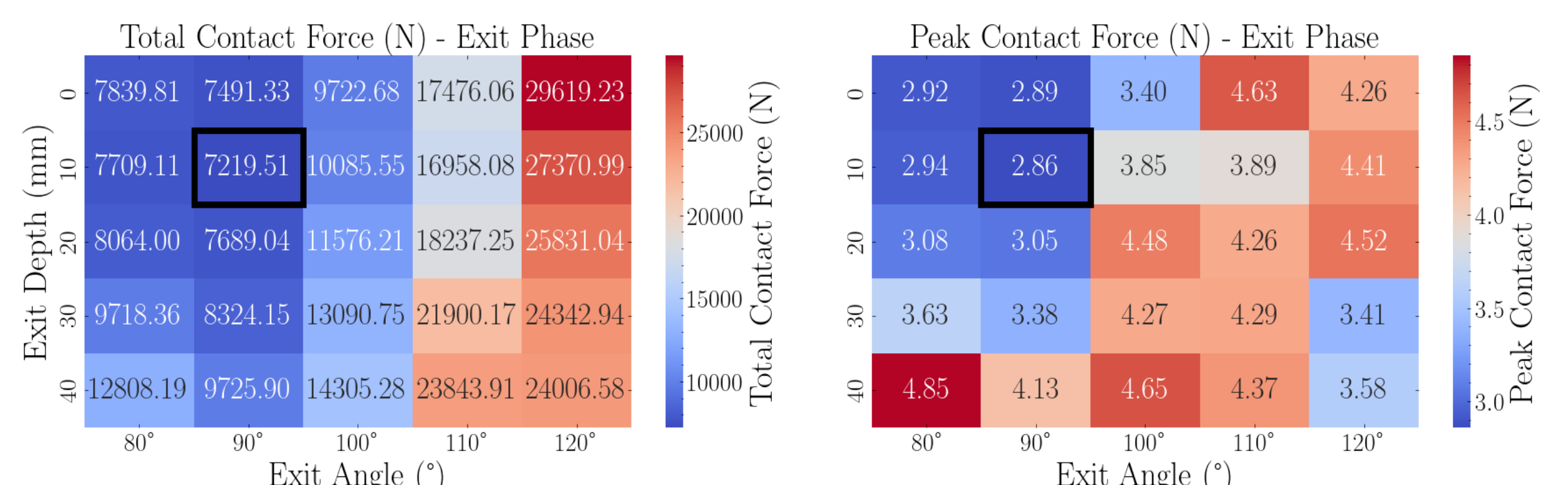


Figure 4: (Left) Total contact force and (Right) Peak contact force during exit phase

5 Conclusion & Future Work

- First integration of soft-body head modeling with realistic bite transfer simulation
- “Straight-in, straight-out” strategy is most comfortable
- Framework provides a safe, scalable tool for testing robot-assisted feeding strategies

Future work

- Add dynamic head and neck motion
- Model tongue, lips, and oral cavity, neck muscles
- Validate contact forces with real users

References:

- [1] Z. Erickson, V. Gangaram, A. Kapusta, C. K. Liu, and C. C. Kemp, “Assistive gym: A physics simulation framework for assistive robotics,” in 2020 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2020, pp. 10 169–10 176.
- [2] R. Ye, W. Xu, H. Fu, R. K. Jenamani, V. Nguyen, C. Lu, K. Dim-itropoulou, and T. Bhattacharjee, “Rcare world: A human-centric simulation world for caregiving robots,” in 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2022, pp.33–40



Paper



Video